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**Allen, IV et al.**

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(54) **SYSTEMS AND METHODS FOR SEALING AND DISSECTING TISSUE USING AN ELECTROSURGICAL FORCEPS INCLUDING A THERMAL CUTTING ELEMENT**

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(51) **Int. Cl.**

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***A61B 18/00*** (2006.01)
***A61B 18/12*** (2006.01)

(52) **U.S. Cl.**

CPC ...... ***A61B 18/1445*** (2013.01); ***A61B 18/1233*** (2013.01); *A61B 2018/00601* (2013.01); *A61B 2018/0063* (2013.01); *A61B 2018/00875* (2013.01); *A61B 2018/00898* (2013.01)

(58) **Field of Classification Search**

None
See application file for complete search history.

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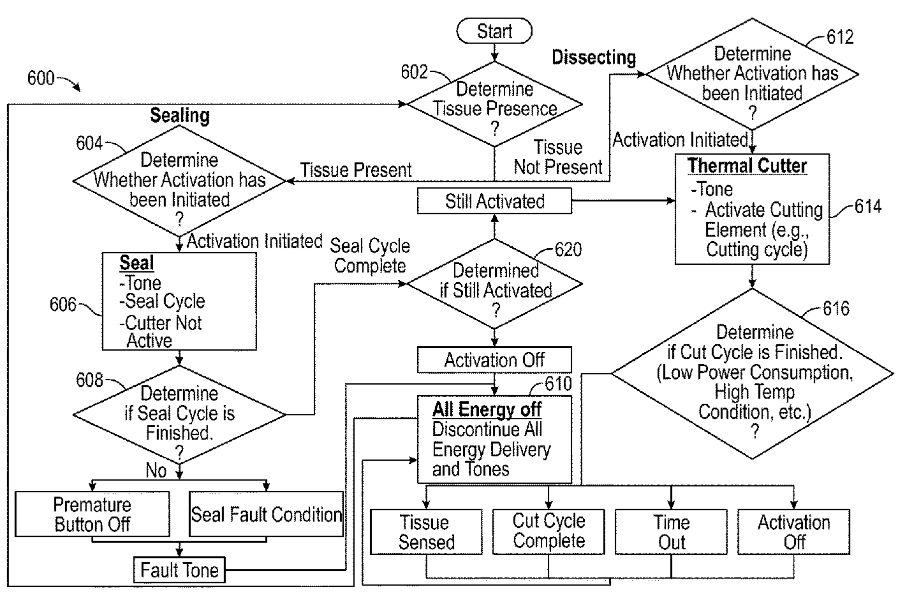
*Assistant Examiner* — Nora W Rhodes

(57) **ABSTRACT**

A method for treating tissue includes determining whether tissue is present between first and second jaw members. In a case where it is determined that tissue is present between the first and second jaw members, the method further includes determining whether activation has been initiated. In a case where it is determined that activation has been initiated, the method further includes supplying electrosurgical energy to the first and second jaw members to seal the tissue and, if it is determined that sealing is complete: stopping the supply of electrosurgical energy; activating a thermal cutting element to supply thermal energy to cut the sealed present tissue; determining if thermal cutting is complete; and, in a case where it is determined that the thermal cutting is complete, stopping and the supply of thermal energy.

**19 Claims, 5 Drawing Sheets**



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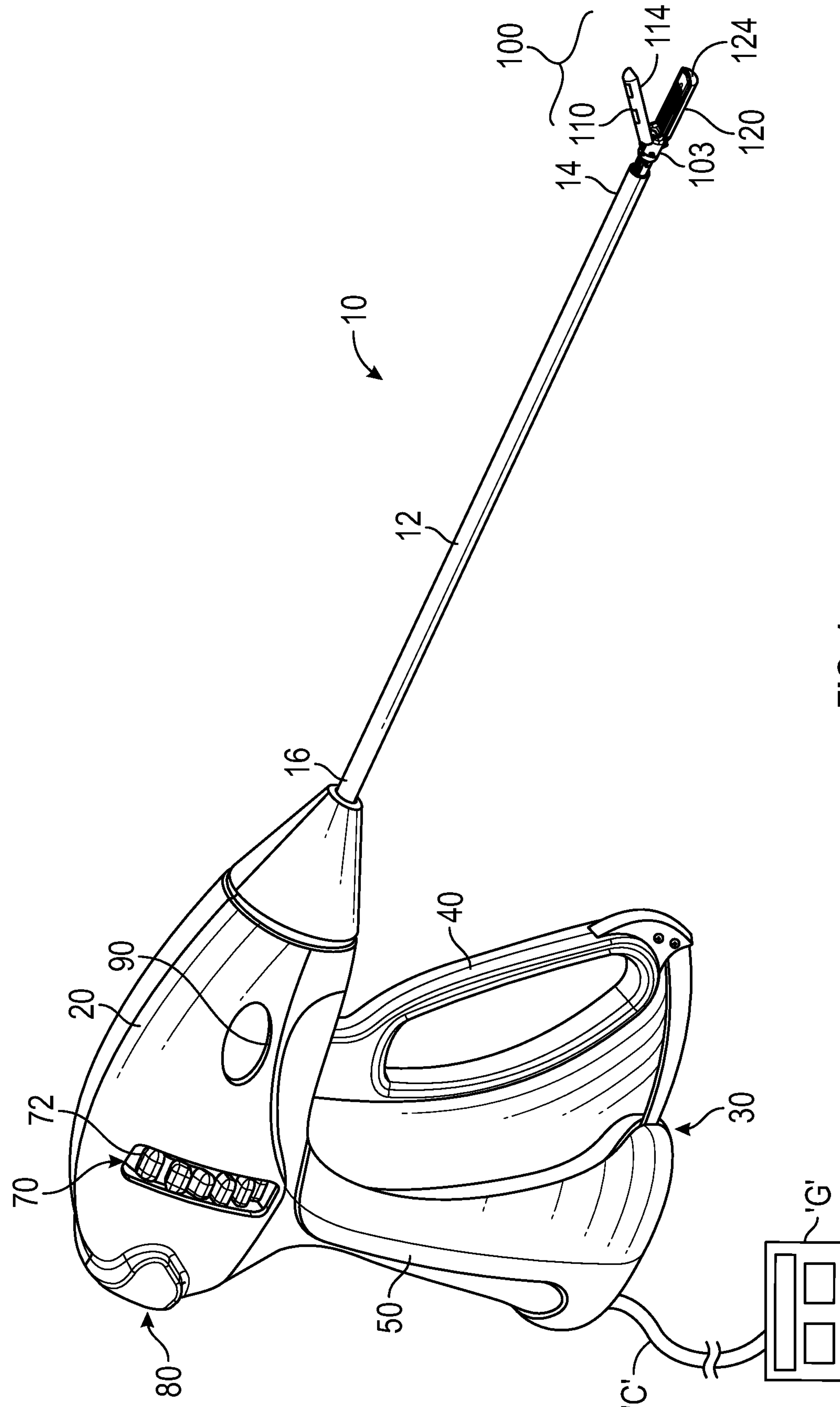


FIG. 1

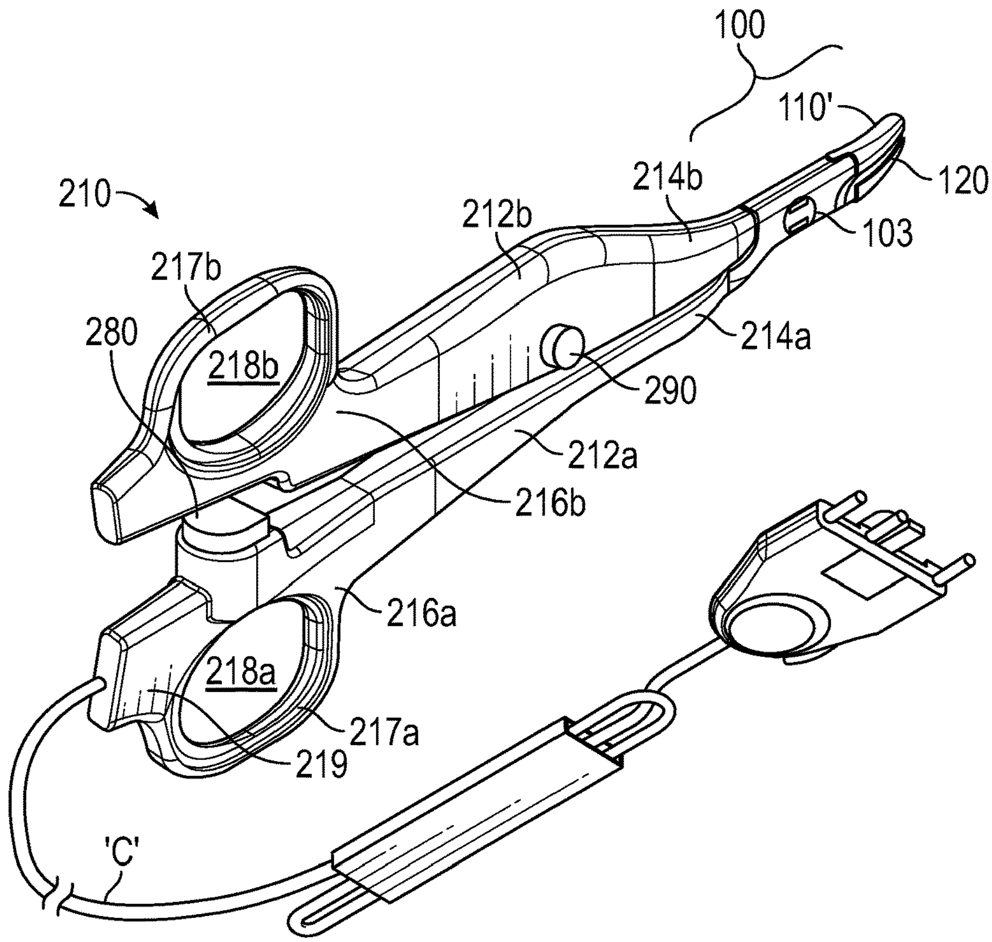
100
110'
120
103
214b
212b
210
217b
218b
280
290
214a
212a
216b
216a
218a
217a
219
'C'

FIG. 2

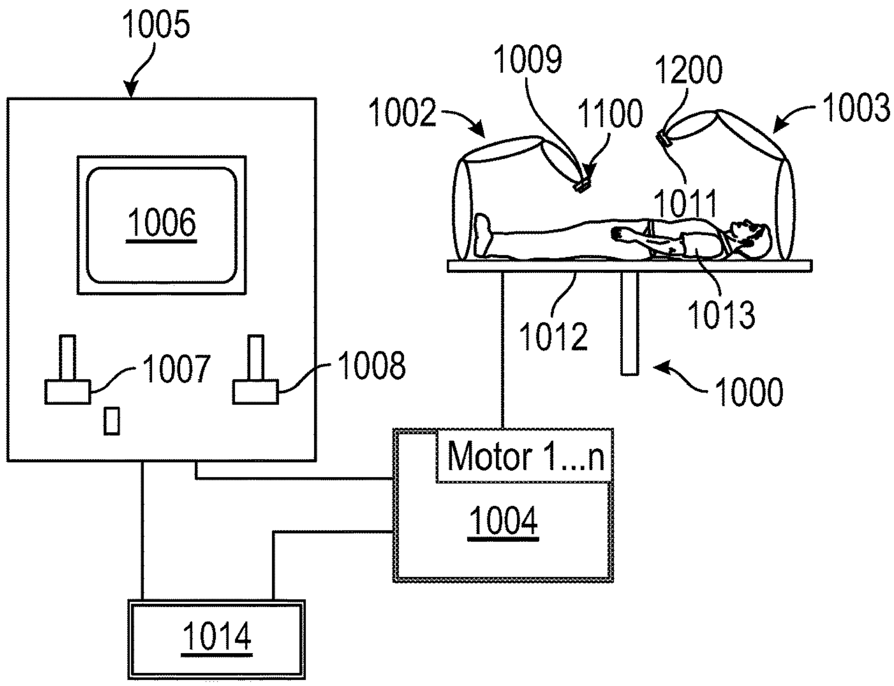
1005
1006
1007
1008
1009
1200
1002
1100
1003
1011
1013
1012
1000
Motor 1...n
1004
1014

FIG. 3

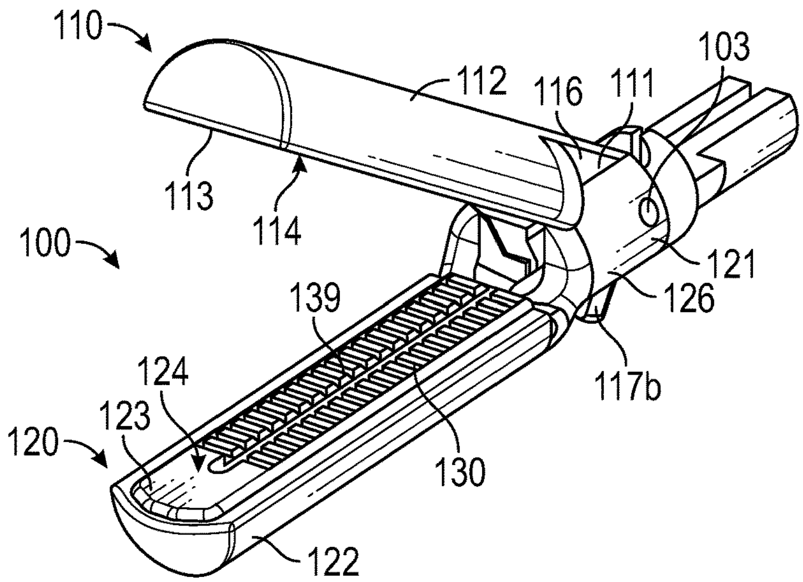


FIG. 4

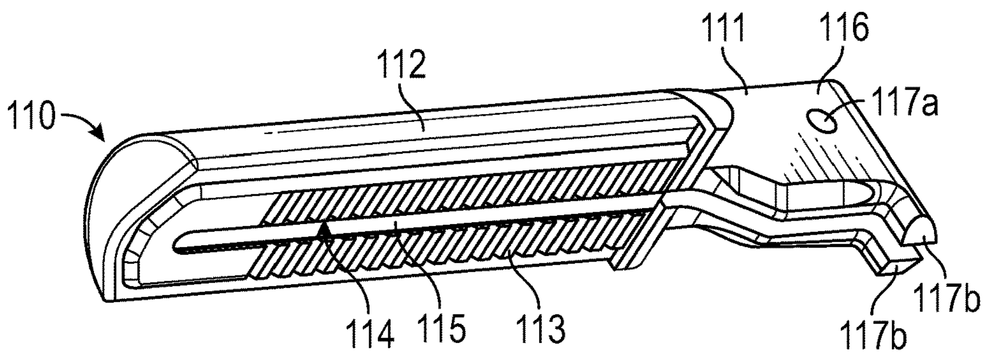


FIG. 5A

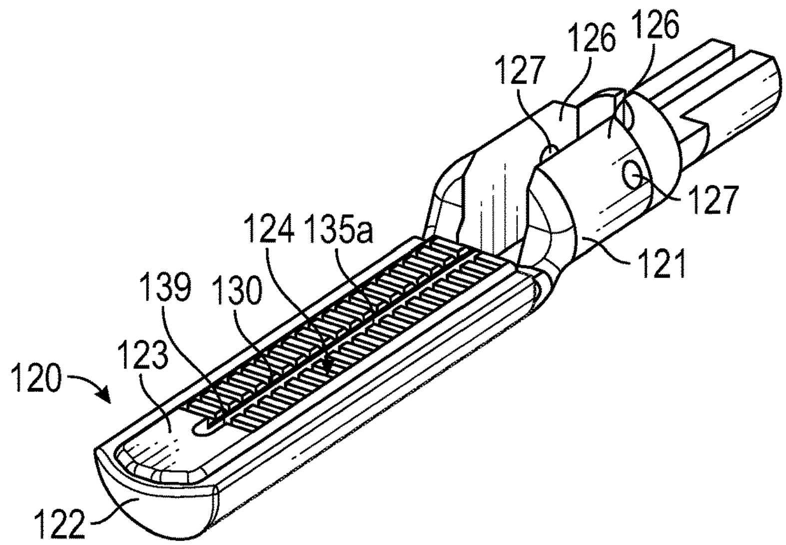


FIG. 5B

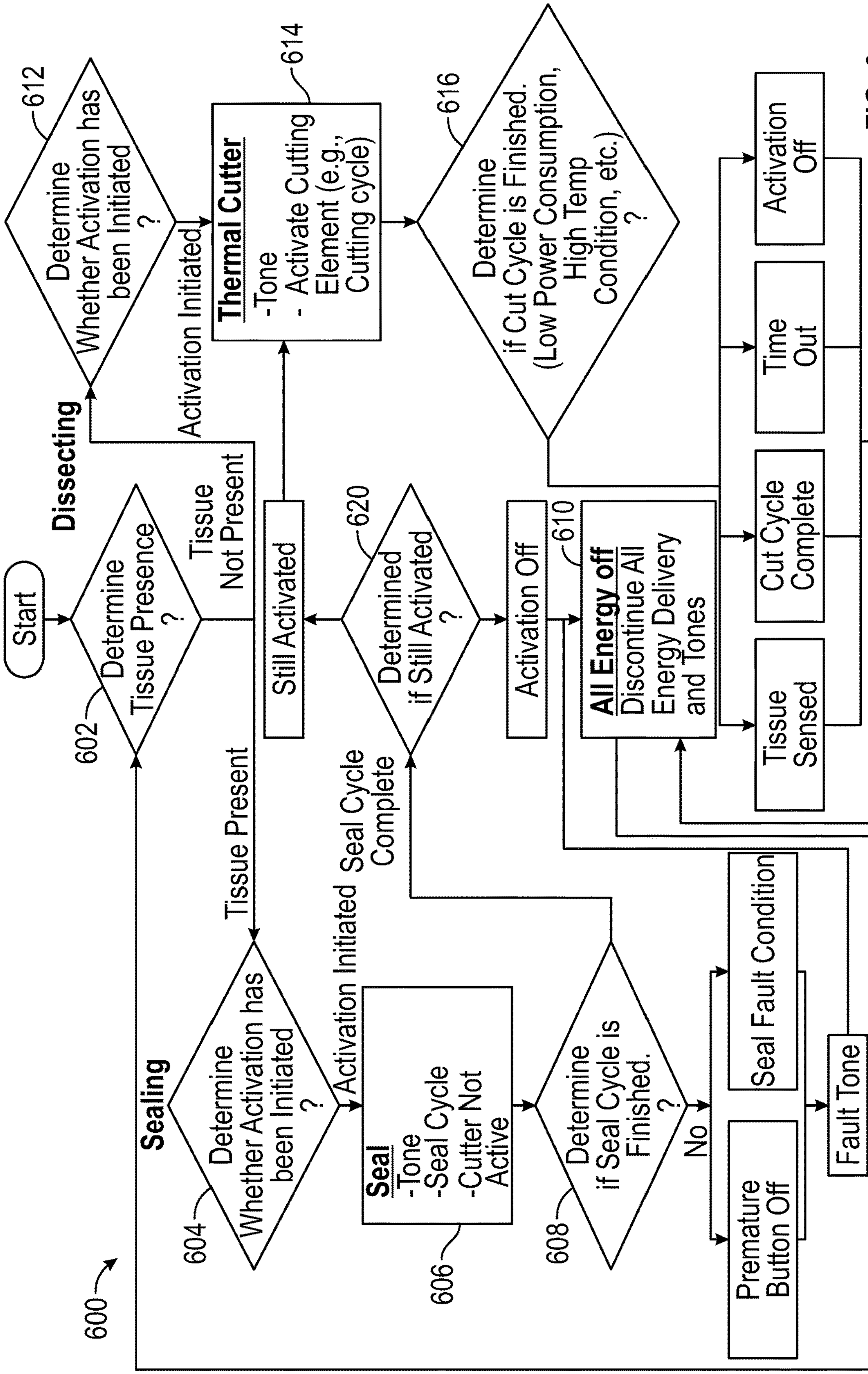
600
Start
Dissecting
Sealing
602
Determine Tissue Presence ?
Tissue Not Present
Tissue Present
612
Determine Whether Activation has been Initiated ?
Activation Initiated
614
Thermal Cutter
-Tone
- Activate Cutting Element (e.g., Cutting cycle)
616
Determine if Cut Cycle is Finished. (Low Power Consumption, High Temp Condition, etc.) ?
604
Determine Whether Activation has been Initiated ?
Activation Initiated
606
Seal
-Tone
-Seal Cycle
-Cutter Not Active
608
Determine if Seal Cycle is Finished. ?
No
Seal Cycle Complete
620
Determined if Still Activated ?
Still Activated
Activation Off
610
All Energy off
Discontinue All Energy Delivery and Tones
Tissue Sensed
Cut Cycle Complete
Time Out
Activation Off
Premature Button Off
Seal Fault Condition
Fault Tone

FIG. 6

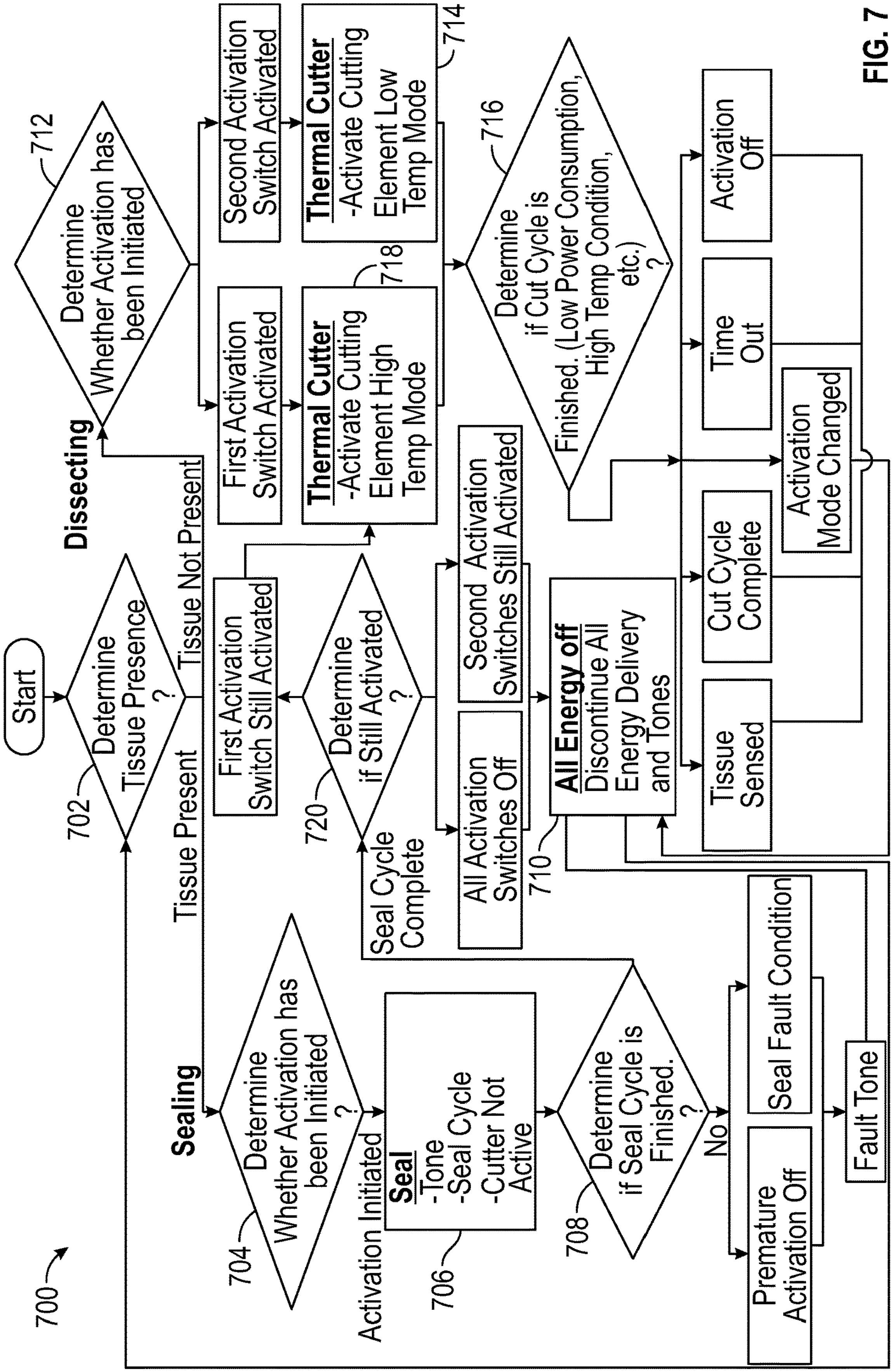
700
Start
Determine Tissue Presence ?
702
Tissue Present
Tissue Not Present
Sealing
Determine Whether Activation has been Initiated ?
704
Activation Initiated
Seal
-Tone
-Seal Cycle
-Cutter Not Active
706
Determine if Seal Cycle is Finished. ?
708
No
Premature Activation Off
Seal Fault Condition
Fault Tone
Seal Cycle Complete
Determine if Still Activated ?
720
All Activation Switches Off
Second Activation Switches Still Activated
First Activation Switch Still Activated
All Energy off
Discontinue All Energy Delivery and Tones
710
Dissecting
Determine Whether Activation has been Initiated
712
First Activation Switch Activated
Second Activation Switch Activated
Thermal Cutter
-Activate Cutting Element High Temp Mode
718
Thermal Cutter
-Activate Cutting Element Low Temp Mode
714
Determine if Cut Cycle is Finished. (Low Power Consumption, High Temp Condition, etc.) ?
716
Tissue Sensed
Cut Cycle Complete
Activation Mode Changed
Time Out
Activation Off

FIG. 7

# SYSTEMS AND METHODS FOR SEALING AND DISSECTING TISSUE USING AN ELECTROSURGICAL FORCEPS INCLUDING A THERMAL CUTTING ELEMENT

## FIELD

The disclosure relates to electrosurgical systems and, more particularly, to systems and methods for sealing and dissecting tissue with an electrosurgical instrument including a thermal cutting element.

## BACKGROUND

A surgical forceps is a pliers-like instrument that relies on mechanical action between its jaw members to grasp, clamp, and constrict tissue. Electrosurgical forceps utilize both mechanical clamping action and energy to heat tissue to treat, e.g., coagulate, cauterize, or seal, tissue. Typically, once tissue is treated, the surgeon has to accurately sever the treated tissue. Accordingly, many electrosurgical forceps are designed to incorporate a knife that is advanced between the jaw members to cut the treated tissue. As an alternative to a mechanical knife, an energy-based tissue cutting element may be provided to cut the treated tissue using energy, e.g., thermal, electrosurgical, ultrasonic, light, or other suitable energy.

## SUMMARY

As used herein, the term "distal" refers to the portion that is being described which is further from a user, while the term "proximal" refers to the portion that is being described which is closer to a user. Further, to the extent consistent, any or all of the aspects detailed herein may be used in conjunction with any or all of the other aspects detailed herein.

Provided in accordance with aspects of the disclosure is a method for treating tissue. The method includes determining whether tissue is present between first and second jaw members and, in a case where it is determined that tissue is present between the first and second jaw members, determining whether activation has been initiated. In a case where it is determined that activation has been initiated, the method includes supplying electrosurgical energy to the first and second jaw members to seal the tissue present between the jaw members and determining if the sealing is complete. In a case where it is determined that sealing is complete, the method further includes: stopping the supply of electrosurgical energy: activating a thermal cutting element to supply thermal energy to cut the sealed tissue; determining if thermal cutting is complete; and, in a case where it is determined that the thermal cutting is complete, stopping the supply of thermal energy.

In an aspect of the present disclosure, activating the thermal cutting element may further include determining if a first thermal mode and/or a second thermal mode is activated. In a case where the first thermal mode is activated, the method includes activating the thermal cutting element in a high temperature mode; and, in a case where the second thermal mode is activated, activating the thermal cutting includes activating the thermal cutting element in a low temperature mode. The thermal cutting element is heated to a lower temperature in the low temperature mode as compared to the high temperature mode.

In another aspect of the present disclosure, the method may further include, in a case where it is determined that tissue is not present between the jaw members: determining whether activation has been initiated; and, in a case where it is determined that activation has been initiated, activating the thermal cutting element to supply thermal energy to tissue to cut the tissue. The method may then include determining if the thermal cutting is complete based on one of power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold. If it is determined that the thermal cutting is complete, the supply of thermal energy is stopped.

In another aspect of the present disclosure, the method may further include, in a case where it is determined that tissue is not present between the jaw members: determining whether activation has been initiated; and, in a case where it is determined that activation has been initiated, activating the thermal cutting element in a different mode to supply thermal energy to tissue to cut the tissue. The different mode is different from a mode of activation of the thermal cutting element to supply thermal energy to cut the sealed tissue. The method may further include determining if the thermal cutting is complete based on one of power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold. In a case where it is determined that the thermal cutting is complete, the supply of thermal energy is stopped.

In still another aspect of the present disclosure, the method further includes, in a case where it is determined that tissue is present between the first and second jaw members, displaying on a display an indication that tissue is present.

In yet another aspect of the present disclosure, the method may further include, in a case where it is determined that tissue is not present between the first and second jaw members, displaying on a display an indication that tissue is not present.

In still yet another aspect of the present disclosure, the method may further include, in a case where it is determined that the sealing is not complete, emitting a fault tone.

In an aspect of the present disclosure, the method may further include displaying on a display a fault condition.

In another aspect of the present disclosure, the method may further include, in a case where the sealing is complete, emitting a success tone.

In an aspect of the present disclosure, determining if the thermal cutting is complete may be based on power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold.

In still yet another aspect of the present disclosure, determining tissue presence between the first and second jaw members may be based on sensing impedance between the first and second jaw members.

Provided in accordance with aspects of the disclosure is a system for treating tissue. The system includes: a surgical instrument, a processor, and a memory coupled to the processor. The surgical instrument includes first and second jaw members configured to treat tissue. The first or second jaw member includes a thermal cutting element and each of the first and second jaw members includes a sealing surface. The memory has instructions stored thereon which, when executed by the processor, cause the system to: determine whether tissue is present between first and second jaw members based on sensing impedance between the sealing surfaces of the first and second jaw members. In a case where it is determined that tissue is present between the first and second jaw members, the system is further caused to determine whether activation has been initiated. In a case where it is determined that activation has been initiated, the system is further caused to supply electrosurgical energy to the first and second jaw members to seal the present tissue and determine if the sealing is complete. In a case where it is determined that the sealing is complete, the system is further caused to: stop the supply of electrosurgical energy: activate a thermal cutting element to supply thermal energy to cut the sealed present tissue: determine if thermal cutting is complete based on power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold; and, in a case where it is determined that the thermal cutting is complete, stop the supply of thermal energy.

In an aspect of the present disclosure, activating the thermal cutting element may further include: determining if a first thermal mode or a second thermal mode is activated: in a case where the first thermal mode is activated, activating the thermal cutting element in a high temperature mode; and in a case where the second thermal mode is activated, activating the thermal cutting element in a low temperature mode, wherein the thermal cutting element is heated to a lower temperature in the low temperature mode as compared to the high temperature mode.

In another aspect of the present disclosure, the instructions, when executed by the processor, may further cause the system to: in a case where it is determined that tissue is not present between the jaw members: determining whether activation has been initiated: in a case where it is determined that activation has been initiated, activating the thermal cutting element to supply thermal energy to tissue to cut the tissue; and determining if thermal cutting is complete based on power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold; and if it is determined that the thermal cutting is complete, stopping the supply of thermal energy.

In another aspect of the present disclosure, the instructions, when executed by the processor, may further cause the system to: in a case where it is determined that tissue is not present between the jaw members: determining whether activation has been initiated: in a case where it is determined that activation has been initiated, activating the thermal cutting element in a different mode to supply thermal energy to tissue to cut the tissue, wherein the different mode is different from a mode of activation of the thermal cutting element to supply thermal energy to cut the sealed tissue; and determining if the different thermal cutting is complete based on power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold; and in a case where it is determined that the thermal cutting is complete, stopping the supply of thermal energy.

In still another aspect of the present disclosure, the instructions, when executed by the processor, may further cause the system to, in a case where it is determined that tissue is present between the first and second jaw members, displaying, on a display, an indication that tissue is present.

In yet another aspect of the present disclosure, the instructions, when executed by the processor, may further cause the system, in a case where it is determined that tissue is not present between the first and second jaw members, to display, on a display, an indication that tissue is not present.

In still yet another aspect of the present disclosure, the instructions, when executed by the processor, may further cause the system, in a case where it is determined that the sealing is not complete, to emit a fault tone.

In an aspect of the present disclosure, the instructions, when executed by the processor, may further cause the system to display, on a display, a fault condition.

Provided in accordance with aspects of the disclosure is a non-transitory storage medium that stores a program causing a computer to execute a method for controlling delivery of tissue dissection with a thermal cutter. The method includes determining whether tissue is present between first and second jaw members and, in a case where it is determined that tissue is present between the first and second jaw members, determining whether activation has been initiated. In a case where it is determined that activation has been initiated, the method further includes supplying electrosurgical energy to the first and second jaw members to seal the tissue present between the jaw members and determining if the sealing is complete. In a case where it is determined that sealing is complete, the method further includes: stopping the supply of electrosurgical energy: activating a thermal cutting element to supply thermal energy to cut the sealed tissue: determining if thermal cutting is complete; and, in a case where it is determined that the thermal cutting is complete, stopping the supply of thermal energy.

## BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects and features of the disclosure will become more apparent in view of the following detailed description when taken in conjunction with the accompanying drawings wherein like reference numerals identify similar or identical elements.

FIG. **1** is a perspective view of a shaft-based electrosurgical forceps provided in accordance with the disclosure shown connected to an electrosurgical generator;

FIG. **2** is a perspective view of a hemostat-style electrosurgical forceps provided in accordance with the present disclosure:

FIG. **3** is a schematic illustration of a robotic surgical system provided in accordance with the present disclosure:

FIG. **4** is a perspective view of a distal end portion of the forceps of FIG. **1**, wherein first and second jaw members of an end effector assembly of the forceps are disposed in a spaced-apart position:

FIG. **5**A is a bottom, perspective view of the first jaw member of the end effector assembly of FIG. **4**:

FIG. **5**B is a top, perspective view of the second jaw member of the end effector assembly of FIG. **4**:

FIG. **6** is a flowchart of a method for tissue sealing and dissection in accordance with aspects of the disclosure; and

FIG. **7** is a flowchart of another method for tissue sealing and dissection in accordance with aspects of the disclosure.

## DETAILED DESCRIPTION

The disclosure relates to electrosurgical systems and methods and, more particularly, to electrosurgical forceps including thermal cutting elements to facilitate tissue treatment, e.g., sealing and/or cutting tissue. Although portions of the disclosure discuss particular types of energy-based surgical systems, the disclosure is equally applicable to other types of energy-based surgical systems not expressly described herein.

Referring to FIG. **1**, a shaft-based electrosurgical forceps provided in accordance with the disclosure is shown generally identified by reference numeral **10**. Aspects and features of forceps **10** not germane to the understanding of the disclosure are omitted to avoid obscuring the aspects and features of the disclosure in unnecessary detail.

Forceps **10** includes a housing **20**, a handle assembly **30**, a rotating assembly **70**, a first activation switch **80**, a second activation switch **90**, and an end effector assembly **100**.

Forceps **10** further includes a shaft **12** having a distal end portion **14** configured to (directly or indirectly) engage end effector assembly **100** and a proximal end portion **16** that (directly or indirectly) engages housing **20**. Forceps **10** also includes cable "C" that connects forceps **10** to an energy source, e.g., an electrosurgical generator "G." Cable "C" includes a wire (or wires) (not shown) extending therethrough that has sufficient length to extend through shaft **12** in order to connect to one or both tissue-treating surfaces **114**, **124** of jaw members **110**, **120**, respectively, of end effector assembly **100** (see FIG. **4**) to provide energy thereto. First activation switch **80** is coupled to tissue-treating surfaces **114**, **124** (FIG. **4**) and the electrosurgical generator "G" for enabling the selective activation of the supply of energy to jaw members **110**, **120** for treating, e.g., cauterizing, coagulating/desiccating, and/or sealing, tissue. Second activation switch **90** is coupled to thermal cutting element **130** of jaw member **120** (FIG. **4**) and the electrosurgical generator "G" for enabling the selective activation of the supply of energy to thermal cutting element **130** of jaw member **120** (FIG. **4**) for thermally cutting tissue. In various embodiments, a multimode and/or a single activation switch may be used.

Handle assembly **30** of forceps **10** includes a fixed handle **50** and a movable handle **40**. Fixed handle **50** is integrally associated with housing **20** and handle **40** is movable relative to fixed handle **50**. Movable handle **40** of handle assembly **30** is operably coupled to a drive assembly (not shown) that, together, mechanically cooperate to impart movement of one or both of jaw members **110**, **120** of end effector assembly **100** about a pivot **103** between a spaced-apart position and an approximated position to grasp tissue between tissue-treating surfaces **114**, **124** of jaw members **110**, **120**. As shown in FIG. **1**, movable handle **40** is initially spaced-apart from fixed handle **50** and, correspondingly, jaw members **110**, **120** of end effector assembly **100** are disposed in the spaced-apart position. Movable handle **40** is depressible from this initial position to a depressed position corresponding to the approximated position of jaw members **110**, **120**. Rotating assembly **70** includes a rotation wheel **72** that is selectively rotatable in either direction to correspondingly rotate end effector assembly **100** relative to housing **20**.

Referring to FIG. **2**, a hemostat-style electrosurgical forceps provided in accordance with the disclosure is shown generally identified by reference numeral **210**. Aspects and features of forceps **210** not germane to the understanding of the disclosure are omitted to avoid obscuring the aspects and features of the disclosure in unnecessary detail.

Forceps **210** includes two elongated shaft members **212***a*, **212***b*, each having a proximal end portion **216***a*, **216***b*, and a distal end portion **214***a*, **214***b*, respectively. Forceps **210** is configured for use with an end effector assembly **100'** similar to end effector assembly **100** (FIG. **4**). More specifically, end effector assembly **100'** includes first and second jaw members **110'**, **120'** attached to respective distal end portions **214***a*, **214***b* of shaft members **212***a*, **212***b*. Jaw members **110'**, **120'** are pivotably connected about a pivot **103'**. Each shaft member **212***a*, **212***b* includes a handle **217***a*, **217***b* disposed at the proximal end portion **216***a*. **216***b* thereof. Each handle **217***a*, **217***b* defines a finger hole **218***a*, **218***b* therethrough for receiving a finger of the user. As can be appreciated, finger holes **218***a*, **218***b* facilitate movement of the shaft members **212***a*, **212***b* relative to one another to, in turn, pivot jaw members **110'**, **120'** from the spaced-apart position, wherein jaw members **110'**, **120'** are disposed in spaced relation relative to one another, to the approximated position, wherein jaw members **110'**, **120'** cooperate to grasp tissue therebetween.

One of the shaft members **212***a*, **212***b* of forceps **210**, e.g., shaft member **212***a*, includes a proximal shaft connector **219** configured to connect forceps **210** to a source of energy, e.g., electrosurgical generator "G" (FIG. **1**). Proximal shaft connector **219** secures a cable "C" to forceps **210** such that the user may selectively supply energy to jaw members **110'**, **120'** for treating tissue. More specifically, a first activation switch **280** is provided for supplying energy to jaw members **110'**, **120'** to treat tissue upon sufficient approximation of shaft members **212***a*, **212***b*, e.g., upon activation of first activation switch **280** via shaft member **212***b*. A second activation switch **290** disposed on either or both of shaft members **212***a*, **212***b* is coupled to the thermal cutting element (not shown, similar to thermal cutting element **130** of jaw member **120** (FIG. **4**)) of one of the jaw members **110'**, **120'** of end effector assembly **100'** and to the electrosurgical generator "G" for enabling the selective activation of the supply of energy to the thermal cutting element for thermally cutting tissue.

Jaw members **110'**, **120'** define a curved configuration wherein each jaw member is similarly curved laterally off of a longitudinal axis of end effector assembly **100'**. However, other suitable curved configurations including curvature towards one of the jaw members **110**, **120'** (and thus away from the other), multiple curves with the same plane, and/or multiple curves within different planes are also contemplated. Jaw members **110**, **120** of end effector assembly **100** (FIG. **1**) may likewise be curved according to any of the configurations noted above or in any other suitable manner.

Referring to FIG. **3**, a robotic surgical system provided in accordance with the disclosure is shown generally identified by reference numeral **1000**. Aspects and features of robotic surgical system **1000** not germane to the understanding of the disclosure are omitted to avoid obscuring the aspects and features of the disclosure in unnecessary detail.

Robotic surgical system **1000** includes a plurality of robot arms **1002**, **1003**: a control device **1004**; and an operating console **1005** coupled with control device **1004**. Operating console **1005** may include a display device **1006**, which may be set up in particular to display three-dimensional images; and manual input devices **1007**, **1008**, by means of which a surgeon may be able to telemanipulate robot arms **1002**, **1003** in a first operating mode. Robotic surgical system **1000** may be configured for use on a patient **1013** lying on a patient table **1012** to be treated in a minimally invasive manner. Robotic surgical system **1000** may further include a database **1014**, in particular coupled to control device **1004**, in which are stored, for example, pre-operative data from patient **1013** and/or anatomical atlases.

The control device **1004** includes a processor connected to a computer-readable storage medium or a memory which may be a volatile type memory, such as RAM, or a non-volatile type memory, such as flash media, disk media, or other types of memory. In various embodiments, the processor may be another type of processor such as, without limitation, a digital signal processor, a microprocessor, an ASIC, a graphics processing unit (GPU), field-programmable gate array (FPGA), or a central processing unit (CPU).

In various embodiments, the memory can be random access memory, read-only memory, magnetic disk memory, solid-state memory, optical disc memory, and/or another type of memory. In various embodiments, the memory can be separate from the control unit and can communicate with the processor through communication buses of a circuit board and/or through communication cables such as serial ATA cables or other types of cables. The memory includes computer-readable instructions that are executable by the processor to operate the control device **1004**. In various embodiments, the control device **1004** may include a network interface to communicate with other computers or a server.

Each of the robot arms **1002**, **1003** may include a plurality of members, which are connected through joints, and an attaching device **1009**, **1011**, to which may be attached, for example, an end effector assembly **1100**, **1200**, respectively. End effector assembly **1100** is similar to end effector assembly **100** (FIG. **4**), although other suitable end effector assemblies for coupling to attaching device **1009** are also contemplated. End effector assembly **1200** may be any end effector assembly, e.g., an endoscopic camera, other surgical tool, etc. Robot arms **1002**, **1003** and end effector assemblies **1100**, **1200** may be driven by electric drives, e.g., motors, that are connected to control device **1004**. Control device **1004** (e.g., a computer) may be configured to activate the motors, in particular by means of a computer program, in such a way that robot arms **1002**, **1003**, their attaching devices **1009**, **1011**, and end effector assemblies **1100**, **1200** execute a desired movement and/or function according to a corresponding input from manual input devices **1007**, **1008**, respectively. Control device **1004** may also be configured in such a way that it regulates the movement of robot arms **1002**, **1003** and/or of the motors.

Turning to FIGS. **4**-**5**B, end effector assembly **100**, as noted above, includes first and second jaw members **110**, **120**. Each jaw member **110**, **120** may include a structural frame **111**, **121**, a jaw housing **112**, **122**, and one or more tissue-treating plates **113**, **123** defining the respective tissue-treating surface **114**, **124** thereof. Alternatively, only one of the jaw members, e.g., jaw member **120**, may include a structural frame **121**, jaw housing **122**, and tissue-treating plate **123** defining the tissue-treating surface **124**. In such embodiments, the other jaw member, e.g., jaw member **110**, may be formed as a single unitary body, e.g., a piece of conductive material acting as the structural frame **111** and jaw housing **112** and defining the tissue-treating surface **114**. An outer surface of the jaw housing **112**, in such embodiments, may be at least partially coated with an insulative material or may remain exposed.

Referring in particular to FIGS. **4** and **5**A, jaw member **110**, as noted above, may be configured similarly as jaw member **120**, may be formed as a single unitary body, or may be formed in any other suitable manner so as to define a tissue-treating surface **114** opposing tissue-treating surface **124** of jaw member **120** and a structural frame **111**. Structural frame **111** includes a proximal flange portion **116** about which jaw member **110** is pivotably coupled to jaw member **120**. In shaft-based or robotic embodiments, proximal flange portion **116** may further include an aperture **117***a* for receipt of pivot **103** and at least one protrusion **117***b* extending therefrom that is configured for receipt within an aperture defined within a drive sleeve of the drive assembly (not shown) such that translation of the drive sleeve, e.g., in response to actuation of movable handle **40** (FIG. **1**) or a robotic drive, pivots jaw member **110** about pivot **103** and relative to jaw member **120** between the spaced-apart position and the approximated position. However, other suitable drive arrangements are also contemplated, e.g., using cam pins and cam slots, a screw-drive mechanism, etc.

Regardless of the particular configuration of jaw member **110**, jaw member **110** further includes a longitudinally-extending insulative member **115** extending along at least a portion of the length of tissue-treating surface **114**. In embodiments, the insulative member **115** may be omitted and the tissue-treating plates **113**, **123** may extend all the way across the jaw member **110**, **120**. Insulative member **115** may be transversely centered on tissue-treating surface **114** or may be offset relative thereto. Further, insulative member **115** may be disposed. e.g., deposited, coated, etc., on tissue-treating surface **114**, may be positioned within a channel or recess defined within tissue-treating surface **114**, or may define any other suitable configuration. Additionally, insulative member **115** may be substantially (within manufacturing, material, and/or use tolerances) coplanar with tissue-treating surface **114**, may protrude from tissue-treating surface **114**, may be recessed relative to tissue-treating surface **114**, or may include different portions that are coplanar, protruding, and/or recessed relative to tissue-treating surface **114**. Insulative member **115** may be formed from, for example, ceramic, parylene, nylon. PTFE, or other suitable material(s) (including combinations of insulative and non-insulative materials).

With reference to FIGS. **4**, and **5**B, as noted above, jaw member **120** includes a structural frame **121**, a jaw housing **122**, and a tissue-treating plate **123** defining the tissue-treating surface **124** thereof. Jaw member **120** further includes a thermal cutting element **130** mounted thereon. Structural frame **121** defines a proximal flange portion **126** and a distal body portion (not shown) extending distally from proximal flange portion **126**. Proximal flange portion **126** is bifurcated to define a pair of flanges having aligned apertures **127** configured for receipt of pivot **103** therethrough to pivotably couple jaw members **110**, **120** with one another. Thermal cutting element **130**, defines an upper, tissue-treating surface **135***a* which may be flat, angled, pointed, curved, or include any suitable configuration. In embodiments where end effector assembly **100**, or a portion thereof, is curved, structural frame **121** and thermal cutting element **130**, or corresponding portions thereof, may similarly be curved, e.g., wherein structural frame **121** and thermal cutting element **130** (or corresponding portions thereof) are relatively configured with reference to an arc (or arcs) of curvature rather than a longitudinal axis. Thus, the terms longitudinal, transverse, and the like as utilized herein are not limited to linear configurations, e.g., along linear axes, but apply equally to curved configurations, e.g., along arcs of curvature. In various embodiments, thermal cutting element **130** may extend all the way to or beyond the distal end of the jaw member **110**, **120**.

Jaw housing **122** may be formed from an electrically insulative material and includes one or more portions (separate or unitary) formed in any suitable manner such as, for example, via overmolding. More specifically, in embodiments, a first overmold may capture structural frame **121** and cutting element **130** while a second overmold captures tissue-treating plate **123**, the first overmold, structural frame **121**, and cutting element **130** to maintain plate **123** isolated from frame **121** and element **130**. Alternatively, only the second overmold may be provided to capture tissue-treating plate **123**, structural frame **121**, and cutting element **130** (while still maintaining the isolation therebetween). As another alternative, an insulative insert may form a portion of jaw housing **122** together with or in place of either or both overmolds. The insulative material, in any of the above configurations, may fill only a portion of the interior of jaw member **120** such that the interior is at least partially hollow, or may fill the substantial entirety of the interior of jaw member **120**. Other suitable configurations are also contemplated.

Regardless of the particular configuration of jaw housing **122**, the assembled jaw member **120** includes tissue-treating plate **123** defining tissue-treating surface **124** and tissue-treating surface **135***a* of thermal cutting element **130** substantially (within manufacturing, material, and/or use tolerances) coplanar with tissue-treating surface **124**, protruding from tissue-treating surface **124**, recessed relative to tissue-treating surface **124**, or provided in any other suitable manner. In aspects, the thermal cutting element **130** may define a variable height, e.g., tapering proximally-to-distally, such that tissue-treating surface **135***a* is angled relative to tissue-treating surface **124**, e.g., to cut tissue heel first. Tissue-treating plate **123**, more specifically, may define a channel **139** through which thermal cutting element **130** at least partially extends such that tissue-treating surface **135***a* thereof is exposed. The remainder of channel **139** may be filled with an insulative material, e.g., a portion jaw housing **122** or other insulator, to isolate thermal cutting element **130** from tissue-treating plate **123**. In the closed position of jaw members **110**, **120**, tissue-treating surface **135***a* of cutting element **130** is configured to oppose insulative member **115** to isolate thermal cutting element **130** from tissue-treating plate **113**.

Generally referring to FIGS. **1**-**5**B, tissue-treating plates **113**, **123** are formed from an electrically conductive material, e.g., for conducting electrical energy therebetween for treating tissue, although tissue-treating plates **113**, **123** may alternatively be configured to conduct any suitable energy, e.g., thermal, microwave, light, ultrasonic, etc., through tissue grasped therebetween for energy-based tissue treatment. As mentioned above, tissue-treating plates **113**, **123** are coupled to activation switch **80** and electrosurgical generator “G” (FIG. **1**) such that energy may be selectively supplied to tissue-treating plates **113**, **123** and conducted therebetween and through tissue disposed between jaw members **110**, **120** to treat tissue, e.g., seal tissue on either side and extending across of thermal cutting element **130**.

Thermal cutting element **130**, on the other hand, is configured to connect to electrosurgical generator “G” (FIG. **1**) and second activation switch **90** to enable selective activation of the supply of energy to thermal cutting element **130** for heating thermal cutting element **130** to thermally cut tissue disposed between jaw members **110**, **120**, e.g., to cut the sealed tissue into first and second sealed tissue portions. Other configurations including multi-mode switches, other separate switches, a single switch for automatic activation, etc. may alternatively be provided.

Thermal cutting element **130** may be any suitable thermal cutting element such as, for example, an aluminum substrate that is Plasma Electrolytic Oxidation (PEO)-treated at least along a portion of tissue-treating surface **135***a*. The thermal cutting element **130** may include a resistive element such that when an AC voltage is applied, resistive element is heated for thermally cutting tissue. As another example, thermal cutting element **130** may be configured as a ferromagnetic (FM) element including a core, e.g., copper, and a ferromagnetic material coated on the core such that when an AC or DC voltage is applied, the FM element is heated up to the Curie point for thermally cutting tissue. Other suitable cutting element configurations are also contemplated. The above-detailed configuration of structural frame **121** of jaw member **120** and thermal cutting element **130**, e.g., wherein there is minimal contact or approximation therebetween (only at the proximal and distal ends of thermal cutting



element **130**) and where free space or insulator is otherwise disposed therebetween, reduces thermal heating of structural frame **121** of jaw member **120** when thermal cutting element **130** is heated (by reducing the conductive pathways for heat to travel to structural frame **121**), thus helping to reduce the overall temperature of jaw member **120** and facilitate cooling after use.

Referring now to FIG. **6**, there is shown a flow diagram of a computer implemented method **600** for tissue sealing and dissection using an electrosurgical end effector assembly including first and second jaw members and a thermal cutting element, e.g., such as the end effector assemblies detailed above. Persons skilled in the art will appreciate that one or more operations of the method **600** may be performed in a different order, repeated, and/or omitted without departing from the scope of the disclosure. In various embodiments, some or all of the operations in the illustrated method **600** can operate using an electrosurgical forceps, e.g., instrument **10** or **210** (see FIGS. **1** and **2**) and the generator “G” (see FIG. **1**), or with robotic surgical system **1000**. Other variations are contemplated to be within the scope of the disclosure. The operations of FIG. **6** will be described with respect to a control device, e.g., control device **1004** of robotic surgical system **1000** (FIG. **3**), a control device incorporated into instrument **10** or **210** (see FIGS. **1** and **2**), a control device incorporated into generator “G” (see FIG. **1**), or any other suitable control device or location thereof including a remotely-disposed control device. It will be understood that the illustrated operations are applicable to other systems and components thereof as well.

Initially at step **602**, the control device determines whether tissue is present between the first and second jaw members. The presence of tissue includes where tissue is between and contacting the sealing surfaces of the first and second jaw members. In embodiments, the control device determines tissue presence between the first and second jaw members based on sensing impedance between the first and second jaw members, e.g., if an impedance is determined, it is determined that tissue is present between the first and second jaw members and if an impedance cannot be determined, it is determined that tissue is not present between the first and second jaw members. In embodiments, the control device determines tissue presence between the first and second jaw members based on a proximity sensor, vision system, and/or jaw aperture. For example, the jaw members may include an imaging device and a processor to detect the presence of tissue. In various embodiments, the control device may determine tissue presence after the activation button is engaged. In various embodiments, when the control device determines that tissue is or is not present between the first and second jaw members, an indication that tissue is or is not present is displayed on a display, e.g., display **1006** (FIG. **3**), a display of generator “G” (FIG. **1**), or other suitable display.

If the control device determines that tissue is present at step **602**, the method proceeds to step **604** where the control device determines whether activation has been initiated. For example, the control device may determine that a first activation switch **80**, **280** (FIGS. **1** and **2**, respectively) is engaged.

Next, at step **606**, when the control device **1004** determines that activation has been initiated, electrosurgical energy is supplied to the first and second jaw members in accordance with a seal cycle algorithm or other suitable algorithm to seal the tissue present, e.g., grasped, between the jaw members. In various embodiments, the control device **1004** may emit, for example a tone, to signal that electrosurgical energy is being supplied to the first and second jaw members.

Next, at step **608**, the control device determines if the seal cycle is complete, e.g., if the tissue has been sealed. In various embodiments, if the seal cycle is complete, the control device emits a success tone. In various embodiments, if the control device determines that the seal cycle is not complete and, thus, that tissue has not been sealed, the control device emits a fault tone. In various embodiments, the control device displays on the display a fault condition. The seal cycle may not complete and, thus, tissue may not be sealed, if, for example, the activation button is prematurely disengaged, due to a fault condition, and/or for other reasons. If the control device determines that the seal cycle is not complete, the method proceeds to step **610** and the supply of electrosurgical energy is ceased.

If it is determined that the seal cycle is complete, the method proceeds to step **620** where the control device determines if the activation switch is still engaged. If the control device determines that the activation switch is still engaged after the seal cycle is complete, the method proceeds to step **614** where the control device emits a tone signaling that dissection will begin. Continuing with step **614**, the control device activates a thermal cutting element in accordance with a thermal cutting cycle (or other suitable algorithm) to supply thermal energy to the thermal cutting element, e.g., thermal cutting element **130** (FIGS. **4** and **5**B) for thermally cutting the previously sealed tissue grasped between the jaw members. At step **616**, the control device determines if the thermal cutting cut cycle is complete. If the thermal cutting cut cycle is determined to be complete, the control device stops the supply of thermal energy at step **610**. In various embodiments, the control device determines if the thermal cutting cut cycle is complete based on power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold. In embodiments, a slow tone may indicate tissue sealing and a fast tone, faster than the slow tone, may indicate cutting tissue.

If the control device determines at step **608** that the first activation switch **80**, **280** is not engaged and the seal cycle is complete, the method proceeds to step **610** where the control device stops the supply of electrosurgical energy.

If at step **602**, the control device determines tissue is not present between the first and second jaw members, the control device determines whether activation has been initiated. For example, the control device may determine that the activation switch is engaged, at step **612**. If the control device determines that activation has been initiated, the control device activates the thermal cutting element in accordance with the thermal cutting cycle to supply thermal energy to tissue to cut the tissue at step **614**, as detailed above. The method them proceed to step **616** and continues as detailed above. In various embodiments, control device **1004** may determine that tissue is not grasped, and the thermal cutter may be activated (i.e., sealing is bypassed). This would allow the clinician to position the cutter against the tissue and transect through the tissue, for example, when minimal hemostasis is required.

Referring now to FIG. **7**, there is shown a flow diagram of a computer implemented method **700** for tissue dissection using an electrosurgical end effector assembly including first and second jaw members and a thermal cutting element, e.g., such as the end effector assemblies detailed above. Persons skilled in the art will appreciate that one or more operations of the method **700** may be performed in a different order, repeated, and/or omitted without departing from the scope of the disclosure. In various embodiments, some or all of the operations in the illustrated method **700** can operate using electrosurgical forceps, e.g., instrument **10** or **210** (see FIGS. **1** and **2**) and the generator "G" (see FIG. **1**) or with robotic surgical system **1000**. Other variations are contemplated to be within the scope of the disclosure. The operations of FIG. **7** will be described with respect to a control device, e.g., control device **1004** of robotic surgical system **1000** (FIG. **3**), a control device incorporated into instrument **10** or **210** (see FIGS. **1** and **2**), a control device incorporated into generator "G" (see FIG. **1**), or any other suitable control device or location thereof including a remotely-disposed control device. It will be understood that the illustrated operations are applicable to other systems and components thereof as well.

Initially at step **702**, the control device determines whether tissue is present between the first and second jaw members. In embodiments, the control device determines tissue presence between the first and second jaw members based on sensing impedance between the first and second jaw members, similarly as noted above. In various embodiments, when the control device determines that tissue is or is not present between the first and second jaw members, an indication that tissue is or is not present is displayed on a display. It is contemplated that the determination of the presence of tissue happens after activation has been initiated, e.g., that the order of steps **704** and **712** occur before step **702** (and similarly with respect to method **600** (FIG. **6**)).

If the control device determines that tissue is present at step **702**, the method proceeds to step **704**, where the control device determines whether activation has been initiated. For example, the control device may determine whether a first activation switch, e.g., switch **80**, **280** (FIGS. **1** and **2**, respectively), and/or a second activation switch, e.g., switch **90**, **290** (FIGS. **1** and **2**, respectively), is engaged. It is contemplated that step **704** could be activated by either the first activation switch, e.g., switch **80**, **280** (FIGS. **1** and **2**, respectively), or the second activation switch, e.g., switch **90**, **290** (FIGS. **1** and **2**, respectively).

If it is determined that activation is initiated at step **706**, e.g., that either or both switches are activated, the method proceeds to step **706**, where electrosurgical energy is supplied to the first and second jaw members in accordance with a seal cycle algorithm or other suitable algorithm to seal the present, e.g., grasped, tissue between the jaw members. In various embodiments, the control device may emit, for example a tone, to signal that electrosurgical energy is being supplied to the first and second jaw members.

Next, at step **708**, the control device determines if the seal cycle is complete. In various embodiments, if the control device determines that the seal cycle is not complete (due to an incomplete seal or disengagement of the first and/or second activation switch, for example), the control device emits a fault tone. In various embodiments, the control device displays on the display a fault condition. In various embodiments, if the seal cycle is determined to be complete and, thus, the tissue is sealed, the control device emits a success tone. In various embodiments, after the seal cycle is complete, the control device determines whether the activation mode has changed from first activation switch to the second activation switch.

If the control device determines at step **708** that the seal cycle is not complete, at step **710** the control device stops the supply of electrosurgical energy.

If the control device determines at step **708** that the seal cycle is complete, the method proceeds to step **720**, where the control device determines if the first or second activation switch is still engaged. If the control device determines that the first activation switch is still engaged after completion of the seal cycle is complete, the method proceeds to step **718** where the control device activates a thermal cutting element in accordance with a first mode, e.g., a high temperature mode, where a suitable thermal cutting cycle or other suitable algorithm is utilized to supply thermal energy to the thermal cutting element for thermally cutting the previously sealed tissue grasped between the jaw members. At step **716**, the control device determines if the thermal cutting cut cycle is complete and, if so, stops the supply of thermal energy at step **710**. In various embodiments, the control device determines if the thermal cutting cycle is complete based on power consumption, temperature exceeding a predetermined threshold, and/or time exceeding a predetermined threshold.

If the control device determines at step **720** that the first activation switch is not engaged but the second activation switch is still engaged after completion of the seal cycle, the method proceeds to step **710** where the control device stops the supply of electrosurgical energy.

Returning to step **702**, if the control device determines that tissue is not present between the first and second jaw members, the control device **1004** then proceeds to step **712** where the control device determines whether activation has been initiated by determining whether the first activation switch or that second activation switch is engaged.

If the control device determines that the first activation switch is engaged, then at step **718** a first thermal mode is activated, activating the thermal cutting element, e.g., in accordance with a high temperature mode of the thermal cutting cycle. If the control device determines that the second activation switch is engaged, then at step **714** a second thermal mode is activated, e.g., activating the thermal cutting element **150** in accordance with a low temperature mode of the thermal cutting cycle, wherein the thermal cutting element is heated to a lower temperature in the low temperature mode as compared to the high temperature mode. Other suitable modes are also contemplated.

Next at step **716**, the control device determines if thermal cutting cycle is complete similarly as detailed above and, if complete, the control device stops the supply of thermal energy at step **710**.

While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

What is claimed is:

**1**. A method, performed by a control device, for treating tissue, comprising:

determining whether an activation switch has been engaged with tissue present between first and second jaw members;

supplying electrosurgical energy to the first and second jaw members to seal the tissue present between the first and second jaw members in response to a determination that the activation switch has been engaged, wherein the electrosurgical energy is conducted between sealing surfaces of the first and second jaw members and through the tissue;

determining if sealing is complete;

stopping the supply of electrosurgical energy in response to a determination that sealing is complete;

determining, in response to a determination that sealing is complete, whether the activation switch is still engaged;

activating a cutting element to supply energy to cut the sealed tissue in response to a determination that the activation switch is still engaged;

determining if cutting is complete; and

stopping the supply of energy in response to a determination that the cutting is complete.

**2**. The method of claim **1**, wherein activating the cutting element includes activating the cutting element in a high temperature mode, and wherein the method further includes:

activating the cutting element in a low temperature mode in response to a determination that the activation switch is not still engaged and that a subsequent activation is initiated.

**3**. The method of claim **1**, further including:

activating the cutting element to supply energy to tissue to cut the tissue in response to a determination that the activation switch has been engaged without tissue present between the first and second jaw members.

**4**. The method of claim **3**, wherein:

the cutting element is activated in a different mode to supply energy to tissue to cut the tissue when the activation switch is engaged without tissue present between the first and second jaw members as compared to a mode wherein the cutting element is activated to supply energy to cut the sealed tissue.

**5**. The method of claim **1**, further including:

determining whether tissue is present between the first and second jaw members; and

displaying on a display an indication that tissue is present in response to a determination that tissue is present between the first and second jaw members.

**6**. The method of claim **5**, further including:

displaying on a display an indication that tissue is not present in response to a determination that tissue is not present between the first and second jaw members.

**7**. The method of claim **1**, further including:

emitting a fault tone in response to a determination that the sealing is not complete.

**8**. The method of claim **7**, further including:

displaying on a display a fault condition.

**9**. The method of claim **1**, further including:

emitting a success tone in response to a determination that the sealing is complete.

**10**. The method of claim **1**, wherein determining whether the activation switch has been engaged with tissue present between the first and second jaw members includes sensing impedance between the first and second jaw members.

**11**. A system for treating tissue, comprising:

a surgical instrument including first and second jaw members configured to treat tissue, at least one of the first or second jaw members including an energizable cutting element, each of the first and second jaw members including a sealing surface;

at least one processor; and

at least one memory coupled to the at least one processor, the at least one memory having instructions stored thereon which, when executed by the at least one processor, cause the system to:

determine whether an activation switch has been engaged with tissue present between the first and second jaw members;

supply electrosurgical energy to the first and second jaw members to seal the tissue present between the first and second jaw members, in response to a determination that the activation switch has been engaged, wherein the electrosurgical energy is conducted between sealing surfaces of the first and second jaw members and through the tissue;

determine if the sealing is complete;

stop the supply of electrosurgical energy in response to a determination that the sealing is complete;

determine, in response to a determination that sealing is complete, whether the activation switch is still engaged;

activate the cutting element to supply energy to cut the sealed tissue in response to a determination that the activation switch is still engaged;

determine if cutting is complete; and

stop the supply of energy in response to a determination that the cutting is complete.

**12**. The system of claim **11**, wherein activating the cutting element includes activating the cutting element in a high temperature mode, and wherein the instructions, when executed by the at least one processor, further cause the system to:

activate the cutting element in a low temperature mode in response to a determination that the activation switch is not still engaged and that a subsequent activation is initiated.

**13**. The system of claim **11**, wherein the instructions, when executed by the at least one processor, further cause the system to:

activate the cutting element to supply energy to tissue to cut the tissue in response to a determination that the activation switch has been engaged without tissue present between the first and second jaw members.

**14**. The system of claim **13**, wherein the instructions, when executed by the at least one processor, further cause the system to:

activate the cutting element in a different mode to supply energy to tissue to cut the tissue when the activation switch is engaged without tissue present between the first and second jaw members as compared to a mode wherein the cutting element is activated to supply energy to cut the sealed tissue.

**15**. The system of claim **11**, wherein the instructions, when executed by the at least one processor, further cause the system to:

determine whether tissue is present between the first and second jaw members; and

display on a display an indication that tissue is present, in response to a determination that tissue is present between the first and second jaw members.

**16**. The system of claim **15**, wherein the instructions, when executed by the at least one processor, further cause the system to:

display on a display an indication that tissue is not present, in response to a determination that tissue is not present between the first and second jaw members.

**17**. The system of claim **11**, wherein the instructions, when executed by the at least one processor, further cause the system to:

emit a fault tone in response to a determination that the sealing is not complete.

**18**. The system of claim **17**, wherein the instructions, when executed by the at least one processor, further cause the system to:

display on a display a fault condition.

**19**. A non-transitory storage medium that stores a program causing a computer to execute a method for treating tissue, the method comprising:

determining whether an activation switch has been engaged with tissue present between first and second jaw members;

supplying electrosurgical energy to first and second jaw members to seal the tissue present between the first and second jaw members in response to a determination that the activation switch has been engaged with tissue present between first and second jaw members, wherein the electrosurgical energy is conducted between sealing surfaces of the first and second jaw members and through the tissue;

determining if sealing is complete;

stopping the supply of electrosurgical energy in response to a determination that sealing is complete;

determining, in response to a determination that sealing is complete, whether the activation switch is still engaged;

activating a cutting element of one of the first or second jaw members to supply energy to cut the sealed tissue in response to a determination that the activation switch is still engaged;

determining if cutting is complete; and

stopping the supply of energy in response to a determination that the cutting is complete.

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